

Ayush Gawai

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Technical Skills

Languages: Python, TypeScript, JavaScript, Java, Rust, SQL, C++, Bash

Backend and Databases: FastAPI, Django, Spring Boot, REST APIs, SQLAlchemy, PostgreSQL, ClickHouse

Frontend: Vue 3, Vite, HTML, CSS, Chart.js

Cloud and Infrastructure: AWS, Docker, Kubernetes, Airflow, Bazel, Helm, ArgoCD, GitHub Actions, Git

AI Integration: LLMs, RAG, Agentic Workflows, PyTorch, scikit-learn

Experience

Software Engineer Intern, Zoox

Jun 2026 – Aug 2026

Tech Stack: Python, TypeScript, FastAPI, PostgreSQL, Databricks, Vue 3, AWS, Airflow, Bazel

- Built and productionized a validation platform for **400+ active autonomous-driving data miners processing 90M+ events weekly**, using existing human labels to continuously measure whether miners returned the intended training data
- Automated miner-specific scoring setup by translating each miner's intent and annotation vocabulary into a verified rule; live-data checks reject invalid logic before scheduled production scoring
- Extended the production service with validation APIs, persisted history, and scheduled fleet sweeps so quality is computed asynchronously and served from stored results without slowing engineer-facing pages
- Built weekly fleet reporting, owner-specific Slack summaries, and event-driven failure alerts, cutting discovery from periodic reviews to **minutes**; write-on-change history reduced redundant validation writes by **93%**

Graduate Research Assistant, San José State University

Aug 2025 – Present

Tech Stack: Python, C++, PyTorch, scikit-learn, OpenCV, NVIDIA Jetson, LeRobot, Isaac Sim, Linux

- Built a reproducible cognitive-workload system that synchronizes eye-tracking and fNIRS streams, extracts temporal features, and evaluates classification signals; the work won the **SJSU RSCA Competition** and advanced to CSU-level presentation
- Built a closed-loop assistive mobility interface that converts AI decisions into haptic feedback and records intervention timing, reaction time, trust, and comfort to evaluate the system's real-time behavior
- Built perception and control prototypes for manipulator, quadruped, and humanoid platforms, integrating vision, simulation, and hardware control to test adaptive robot behavior

Software Engineer, Zscaler

Jan 2024 – Aug 2025

Tech Stack: Python, SQL, Snowflake, dbt, Matillion, AWS, Docker, GitHub Actions, REST APIs

- Built a metadata-driven service that resolves nested Salesforce formula dependencies and converts them into validated, version-controlled SQL, cutting reporting latency by approximately **80%**
- Reworked full-table refreshes into incremental change processing and optimized execution, improving query performance by approximately **35%** while reducing warehouse compute overhead
- Integrated external business systems into normalized production workflows and automated reconciliation checks, reducing manual investigation when source logic or API schemas changed
- Containerized jobs and added schema and SQL checks in CI to standardize environments and catch deployment issues earlier

Software Engineer Intern, Coditas

Feb 2021 – Aug 2021

Tech Stack: Java, Spring Boot, PostgreSQL, Liquibase, Docker, GitLab CI

- Improved production microservice reliability through concurrency, connection-pooling, caching, and database-query fixes, reducing request failures by approximately **33%** and improving query performance by **2.5x**

Education

San José State University, M.S. Applied Data Science (GPA: 3.8)

Aug 2025 – May 2027

Coursework: Data Structures and Algorithms, Distributed Systems, Software Engineering, Databases, Cloud Computing, AI, ML

Projects

CarbonShift | Python, REST APIs, Agentic Workflows, LLMs

- Built a real-time compute orchestration system that shifts workloads using live carbon-intensity and electricity-price signals, demonstrating **43% simulated carbon reduction** and **37% cost optimization**

Transit Analytics | Python, SQL, Snowflake, REST APIs, Geospatial Data

- Built a near-real-time telemetry system from API ingestion through anomaly handling and route-level analysis, converting noisy GPS streams into operational performance insights

Sneakstore.in | E-commerce, REST APIs, Webhooks, Python, SQL

- Built and operated an e-commerce platform end to end, owning the storefront, inventory flows, backend integrations, analytics, hosting, and deployment
- Integrated payment webhooks and logistics APIs to automate order confirmation and shipment tracking; used funnel and pricing analysis to guide product decisions

Achievements

- **RSCA Fellowship:** Awarded by the SJSU Office of Research for original applied AI and systems research
- **RSCA Research Competition Winner:** Selected as SJSU delegate to the CSU-level research competition across disciplines
- **Global Spartan Scholarship:** Merit-based SJSU award recognizing academic excellence and university contribution
- **Student Hiring Committee Member:** Evaluated faculty candidates and contributed structured student feedback at SJSU